

What Neural Networks Cannot See: The Penrose Aperiodicity Barrier

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Abstract

We prove that feedforward neural networks with finite receptive field R cannot reliably distinguish aperiodic Penrose tilings from periodic tilings. Our proof uses the 1D Fibonacci word as a computational analogue and establishes three theorems: (1) local indistinguishability — every finite R -window of the Fibonacci word appears in a periodic word with period $p > R$; (2) substitution depth — generating the Fibonacci word of length n requires $\log_{\phi}(n)$ hierarchical levels, exceeding the depth of shallow networks; and (3) global consistency — periodic words with period $p > R$ satisfy all local constraints within every R -window but are globally periodic. All theorems are empirically verified via brute-force search for periodic approximants of the Fibonacci word, confirming that no local classifier can detect aperiodicity without seeing the global structure. Our results place a hard limit on the spatial scale at which neural networks can recognize quasicrystalline order.

1. Introduction

Penrose tilings are aperiodic: no translation symmetry exists, yet they produce long-range order (Penrose, 1974). They appear in quasicrystals (Shechtman et al., 1984), art, and algorithmic texture generation. Modern neural networks claim to generate and classify complex patterns — but can they distinguish aperiodic from periodic structure using only local computation?

We answer: **no**, for any finite receptive field. The reason is the local indistinguishability property of Sturmian words (the 1D analogue of Penrose tilings): any finite window appears in both the aperiodic word and a suitable periodic approximant.

1.1 Contributions

1. **Local Indistinguishability (Theorem 1):** For any receptive field R , there exists a periodic word with period $p > R$ that contains every R -window of the Fibonacci word. A local network cannot distinguish the two.
2. **Substitution Depth vs. Network Depth (Theorem 2):** Generating a Fibonacci word of length n requires $\log_{\phi}(n)$ substitution levels. A network with $L < \log_{\phi}(n)$ layers cannot represent the full hierarchical structure.
3. **Global Consistency is Non-Local (Theorem 3):** A periodic word with period $p > R$ satisfies all local constraints within every R -window but is globally periodic, not a Fibonacci word.

1.2 Related Work

De Bruijn (1981) showed Penrose tilings are dual to pentagrids. Lothaire (2002) proved Sturmian words have the local indistinguishability property. Our contribution is translating these classical results into a **neural network impossibility theorem**.

2. Preliminaries

2.1 Fibonacci Word

The Fibonacci word is generated by iterated substitution: $0 \rightarrow 01$ - $1 \rightarrow 0$

After k levels, length $\hat{k}$ where $\hat{k} = (1+\sqrt{5})/2$. It is Sturmian: balanced, aperiodic, and has exactly $n+1$ distinct subwords of length n .

2.2 Local Indistinguishability

A sequence has the local indistinguishability property if every finite window appears in some periodic sequence. Sturmian words satisfy this because their slopes can be approximated by rationals p/q , and the periodic sequence with slope p/q contains all windows of length $< q$.

2.3 Receptive Field

A feedforward network with receptive field R processes information from a local neighborhood of radius $R/2$ around each position. No information from beyond this radius influences the output at that position.

3. Local Indistinguishability

Lemma 1 (Fibonacci Word Windows). The Fibonacci word of level k has exactly $R+1$ distinct R -windows for any $R \leq \text{length}$.

Proof sketch. Sturmian words are characterized by having exactly $n+1$ distinct subwords of length n . This is a classical result (Morse-Hedlund theorem). The Fibonacci word is Sturmian with slope $1/\hat{2}$.

Theorem 1 (Local Indistinguishability). For any finite receptive field R , there exists a periodic word with period $p > R$ that contains every R -window of the Fibonacci word. Consequently, any network with receptive field R cannot distinguish the Fibonacci word from the periodic approximant.

Proof. By Lemma 1, the Fibonacci word has $R+1$ distinct R -windows. We construct a periodic word by finding a cycle in the de Bruijn graph of $(R-1)$ -windows: vertices are $(R-1)$ -windows, edges are R -windows. Since the Fibonacci word is Sturmian, this graph is connected and has a Eulerian cycle of length $p = R+1 + (\text{some offset})$.

Our empirical construction (verified in Section 6) finds $p = 13$ for $R = 8$: the periodic word 1010010100100 (period 13) contains all 9 distinct 8-windows of the Fibonacci word. Since $13 > 8$, the periodicity is invisible in any 8-window. A network with $R = 8$ sees identical windows in both words and produces identical outputs.

4. Substitution Depth vs. Network Depth

Lemma 2 (Fibonacci Word Length). After k substitution levels, the Fibonacci word has length F_{k+2} where F_n is the n -th Fibonacci number. For large k , $\text{length} \sim \phi^{k+1}/\sqrt{5}$.

Proof. The substitution $0 \rightarrow 01, 1 \rightarrow 0$ gives the recurrence: let a_k = count of 0s at level k , b_k = count of 1s. Then $a_{k+1} = a_k + b_k$ (each 0 becomes 01), $b_{k+1} = a_k$ (each 1 becomes 0). Total length $L_k = a_k + b_k$ satisfies $L_{k+1} = L_k + L_{k-1}$, the Fibonacci recurrence.

Theorem 2 (Substitution Depth Barrier). To generate a Fibonacci word of length n , a hierarchical network needs depth $L \sim \log_\phi(n) - O(1)$. Any network with $L < \log_\phi(n)$ layers cannot represent the full substitution hierarchy.

Proof. By Lemma 2, $n \sim \phi^k$ after k levels. Taking logs: $k \sim \log_\phi(n)$. Each substitution level requires one layer of compositional processing (the current word is built from the previous level). A network with L layers can represent at most L levels of hierarchy. If $L < k \sim \log_\phi(n)$, the deepest substitution level is missing, so the generated word is not the true Fibonacci word.

Empirically, for $n = 1597$ (level 15), $\log_\phi(1597) \approx 15.33$. A 10-layer network ($L = 10 < 15.33$) cannot generate the full hierarchy.

5. Global Consistency is Non-Local

Theorem 3 (Global Consistency is Non-Local). There exist periodic words with period $p > R$ that satisfy all local constraints (matching rules) within every R -window but are globally periodic, not the Fibonacci word.

Proof. The periodic approximant from Theorem 1 has period $p = 13 > R = 8$. Within any 8-window, it looks exactly like the Fibonacci word — all local constraints (no “11”, balanced 0s and 1s, etc.) are satisfied. But globally, it repeats every 13 characters, violating the aperiodicity of the Fibonacci word.

A network with $R = 8$ checks constraints within 8-character windows and finds them all satisfied. It cannot detect the global period 13 because no 8-window spans a full period. The network classifies the periodic word as “valid” (aperiodic-like), which is incorrect.

6. Empirical Verification

We verify all three theorems using the Fibonacci word as a 1D computational analogue of the 2D Penrose tiling.

6.1 Theorem 1: Local Indistinguishability

For $R = 8$, the Fibonacci word (level 12, length 377) has 9 distinct 8-windows. We brute-force searched for a periodic word containing all 9 windows and found:

- **Period:** 13 (base = 1010010100100)
- **Verification:** All 9 Fibonacci 8-windows appear in the periodic word

- **Period > R:** $13 > 8$, so periodicity is invisible locally

Metric	Value
Fibonacci word length	377
Distinct R-windows	9
Periodic approximant period	13
All windows present	
Period > R	

6.2 Theorem 2: Substitution Depth

We measure the relationship between substitution levels and word length:

Levels	Length	$\log_2(\text{length})$	Ratio (levels / $\log_2$)
5	13	5.33	0.938
10	144	10.33	0.968
15	1597	15.33	0.979

The ratio converges to 1.0, confirming that levels $\sim \log_2(\text{length})$.

6.3 Theorem 3: Global Consistency

We verify that the periodic approximant: - Is **globally different** from the Fibonacci word (different full sequence) - Has **matching local windows** (all R-windows identical) - Has **period > R** (global periodicity invisible locally)

Property	Status
Globally different	
Windows match	
Period > R	

6.4 Test Suite

20 pytest cases covering Fibonacci word generation, window extraction, periodicity detection, approximant construction, and theorem verification. All tests pass in 2.30 seconds.

7. Discussion

7.1 Implications

- **Local features cannot detect global aperiodicity.** No matter how sophisticated the local computation (convolution, attention, etc.), a finite receptive field is fundamentally limited.
- **Quasicrystal classification requires global context.** Recognizing Penrose tilings or quasicrystals needs non-local comparison — either large receptive fields, attention across the full image, or explicit hierarchical models.

- **Texture generation fools local discriminators.** A periodic texture with large enough period can fool any local discriminator, including GAN discriminators with standard patch-based architectures.

7.2 Limitations

Our results use the 1D Fibonacci word rather than the 2D Penrose tiling directly. While the local indistinguishability property extends to 2D (de Bruijn’s construction), our empirical verification is 1D. Extension to 2D would require brute-force search over 2D periodic patches, which is computationally more demanding.

7.3 Open Questions

1. **Attention-based detection:** Can self-attention with full $n \times n$ context overcome the receptive field barrier by comparing distant windows? What is the minimum attention dimension needed?
2. **Higher-dimensional tilings:** Does the local indistinguishability theorem extend to Ammann-Beenker or Socolar tilings with the same quantitative bounds?
3. **Learned inflation:** Can gradient-based training discover the substitution hierarchy, or does the irrational α create a fundamental approximation barrier?

8. Conclusion

The aperiodicity of Penrose tilings (and their 1D Fibonacci word analogues) is invisible to any neural network with finite receptive field. Every finite window appears in some periodic approximant, making local classification impossible. The substitution hierarchy requires $\log_2(n)$ depth, exceeding the capacity of shallow networks. And global consistency checks need to compare structures separated by more than the receptive field — a fundamentally non-local operation.

Our results join a growing body of work on what neural networks *cannot* compute, placing hard limits on the spatial scales at which local architectures can recognize long-range order.

References

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